Classification and Regression of ECG, EEG, and Time-Series Signals

Project Report for Digital Signal Processing (CSE430)

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April 25, 2026

Abstract

This report presents a comparative study of machine learning and deep learning approaches for biomedical and time-series signal analysis using four datasets: *AbnormalHeartBeat*, *SelfRegulationSCP2*, *Appliances Energy*, and *LiveFuelMoisture-Content*. The project was completed in two phases. In Phase 1, pre-extracted tabular features were used to train Ridge models and Dense Neural Networks. In Phase 2, the same datasets were provided as raw tensors with shape $(\text{batch}, \text{channel}, \text{length})$, and deep sequence models including CNN, Vanilla RNN, LSTM, and GRU were trained directly on the raw inputs. The report summarizes the datasets, experimental setup, exploratory findings, performance across models, and a final comparison between feature-based learning and raw-signal learning.

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1 Introduction

Signal analysis plays a major role in modern data-driven decision making, especially for biomedical and sensor-based time-series data. ECG and EEG signals contain structured temporal information that can be used for classification, while multivariate sensor streams can support regression tasks. In this project, I explored both classical machine learning and deep learning methods to understand how model performance changes when learning from extracted features versus raw sequential signals.

The project was completed as part of the **Digital Signal Processing (CSE430)** course at **BRAC University**. It was divided into two phases. Phase 1 focused on tabular feature learning, while Phase 2 focused on representation learning from raw data.

2 Project Objectives

The major goals of the project were as follows:

- Compare linear and nonlinear models on extracted features.
- Compare CNN and recurrent models on raw 3D signal tensors.
- Analyze the characteristics of the four datasets, including shape, task type, channels, class balance, and value range.
- Determine which models performed best on classification and regression tasks and explain the likely reasons.

3 Project Structure

3.1 Phase 1: Learning from Extracted Features

In Phase 1, each dataset was provided in tabular form, where each row represented one sample and each column represented an extracted feature. Since signal descriptors had already been computed, the learning problem focused on mapping those features to class labels or continuous targets. The following models were used:

- Ridge Classifier for classification datasets
- Ridge Regression for regression datasets
- Dense Neural Network for both task types

This phase was important for establishing baselines and for testing whether engineered features were already sufficient for strong performance.

3.2 Phase 2: Learning from Raw Signals

In Phase 2, the same datasets were provided in raw tensor form with shape $(\text{batch}, \text{channel}, \text{length})$. The models therefore learned temporal and channel-level representations directly from the raw input. The main architectures were:

- Convolutional Neural Network (CNN)
- Vanilla RNN
- LSTM
- GRU

This phase was more challenging because the model had to learn both informative features and predictive structure at the same time.

4 Dataset Overview

The project used four datasets. The first two datasets, *AbnormalHeartBeat* and *SelfRegulationSCP2*, were used for classification. The last two datasets, *Appliances Energy* and *LiveFuelMoistureContent*, were used for regression.

For compact presentation in the tables, the following abbreviations are used: AHB = *AbnormalHeartBeat*, SCP2 = *SelfRegulationSCP2*, AE = *Appliances Energy*, and LFMC = *LiveFuelMoistureContent*.

4.1 Phase 1 Datasets

Table 1: Classification datasets used in Phase 1

			Shape		**Classes**	
Dataset	**Signal**	**Task**	**Train**	**Test**	**Count**	**Labels**
AHB	ECG	Class.	205×20000	204×20000	2	0, 1
SCP2	EEG	Class.	180×20000	200×20000	2	0, 1

Table 2: Regression datasets used in Phase 1

			Shape		**Target Statistics**	
Dataset	**Signal**	**Task**	**Train**	**Test**	**Mean**	**Variance**
AE	Time Series	Regr.	95×20000	42×20000	14.165	18.403
LFMC	Time Series	Regr.	3493×20000	1510×20000	117.013	1727.951

4.2 Phase 2 Tensor Structure

Table 3: Classification tensors used in Phase 2

			Batch		Input	
Dataset	**Signal**	**Task**	**Train**	**Test**	**Channels**	**Length**
AHB	ECG	Class.	204	205	1	18530
SCP2	EEG	Class.	200	180	7	1152

Table 4: Regression tensors used in Phase 2

			Batch		Input	
Dataset	**Signal**	**Task**	**Train**	**Test**	**Channels**	**Length**
AE	Time Series	Regr.	95	42	24	144
LFMC	Time Series	Regr.	3493	1510	7	365

5 Exploratory Data Analysis

5.1 Classification Datasets

The two classification datasets show different modeling challenges. *AbnormalHeartBeat* is a small ECG dataset with only one channel but a very long sequence length of 18,530. Its class distribution is imbalanced, with 149 samples in one class and 55 in the other, which makes optimization more sensitive to biased decision boundaries. Because the value range is relatively narrow, explicit normalization is less critical here than class-aware training strategies such as weighted loss.

In contrast, *SelfRegulationSCP2* is a multi-channel EEG dataset with 7 channels and a shorter sequence length of 1,152. The class distribution is balanced, which is helpful for classification, but the signal range varies widely across channels. For this reason, per-channel standardization is important before training deep models. Compared with the ECG dataset, this EEG dataset is structurally richer but also more dependent on proper scaling and channel-aware representation learning.

5.2 Regression Datasets

The regression datasets mainly differ in dataset size and feature scale. *Appliances Energy* is a very small multivariate dataset with 24 channels and sequence length 144. Although its target distribution is close to normal, the limited number of training samples makes

the task sensitive to overfitting and unstable generalization. Since each channel operates over a different numeric range, normalization is necessary before model training.

LiveFuelMoistureContent is much larger, with 3,493 training samples and 1,510 test samples, which makes it better suited for deep learning than *Appliances Energy*. However, the raw values span a much wider range, reaching up to 32,767 in several channels. This makes both input normalization and target scaling important. Overall, the regression EDA suggests that dataset scale and normalization are the two most important factors affecting performance.

5.3 Main Observations

- Long sequences favor models that can preserve temporal information over many steps.
- Class imbalance makes the ECG classification task harder than raw accuracy alone may suggest.
- Multi-channel EEG and sensor data require careful scaling because channel ranges differ significantly.
- Small-sample regression is vulnerable to overfitting even when the target distribution is well behaved.

6 Methodology

6.1 Phase 1 Models

Ridge Classifier and Ridge Regression. Ridge models are linear methods with L_2 regularization. They are fast, stable, and effective when extracted features already contain strong predictive information.

Dense Neural Network. The Dense Neural Network was used to test whether nonlinear transformations of the extracted features could improve performance over linear models.

6.2 Phase 2 Models

CNN. CNN was used to learn local temporal patterns and compact feature hierarchies from the raw sequences.

Vanilla RNN. Vanilla RNN was used as a simple sequential baseline, but it is more vulnerable to vanishing gradients and long-sequence difficulties.

LSTM and GRU. These recurrent variants were used because gated architectures can better preserve useful sequential information and are generally more effective on longer or

more complex series.

6.3 Model Architectures and Hyperparameters

The experimental setup used dataset-specific CNN and recurrent configurations. The CNN models differed mainly in convolution depth, feature width, pooling schedule, and fully connected head size. The recurrent models differed in hidden size, number of layers, whether a CNN-based downsampling stage was used before the recurrent block, and the classifier or regressor head.

Table 5: CNN architectures and estimated trainable parameters

	Convolution		Head				
Dataset	**Channels**	**Kernels / Pools**	**FC Sizes**	**Dropout**	**Output**	**Task**	**Parameters**
AHB	4,8,8,16	5,5,5,5 / 5,5,3	256,128	0.35	2	Class.	1,045,402
SCP2	16,32,64	5,5,5 / 3,3,3	256,128	0.50	2	Class.	734,626
AE	32,64,128	5,5,5 / 3,3,3	128,64	0.20	1	Regr.	145,633
LFMC	16,32,64	5,5,5 / 3,3,3	128,64	0.20	1	Regr.	127,793

Table 6: Recurrent architectures and estimated trainable parameters

		Recurrent Core			Parameters		
Dataset	**Pre-RNN Processing**	**Hidden**	**Layers**	**FC Head**	**RNN**	**LSTM**	**GRU**
AHB	CNN downsampling: ch 4,8; pool 5,5	16	3	none	1,834	6,042	4,802
SCP2	No pre-downsampling	256	3	128	364,162	1,356,930	1,026,178
AE	No pre-downsampling	128	3	128	102,401	359,681	273,921
LFMC	No pre-downsampling	128	3	128	100,225	350,977	267,393

Parameter-count note. The parameter counts above are estimated from the reported layer sizes using standard 1D convolution, linear, RNN, LSTM, and GRU formulas with bias terms included. For classification datasets, the output layer size is 2; for regression datasets, the output layer size is 1.

7 Results

7.1 Phase 1 Results

7.1.1 Classification Results

Table 7: Phase 1 classification performance with extracted features

		Train		Test	
Dataset	**Model**	**Loss**	**Accuracy**	**Loss**	**Accuracy**
AHB	Ridge Classifier	—	96.5%	—	66.8%
AHB	Dense NN	1.1614	73.04%	1.2130	73.17%
SCP2	Ridge Classifier	—	99.0%	—	51.6%
SCP2	Dense NN	1.6862	50.00%	1.7326	50.00%

7.1.2 Regression Results

Table 8: Phase 1 regression performance with extracted features

		Train		Test	
Dataset	**Model**	**Loss**	**NRMSE**	**Loss**	**NRMSE**
AE	Ridge Regression	—	0.7390	—	1.1500
AE	Dense NN	26.3625	0.7285	34.5957	1.2106
LFMC	Ridge Regression	—	0.8670	—	1.1000
LFMC	Dense NN	1641.7976	0.9465	1897.6979	1.0320

7.2 Phase 2 Results

7.2.1 Classification Results on Raw Signals

Table 9: Phase 2 classification performance on raw tensors

		Performance	
Dataset	**Model**	**Loss**	**Accuracy**
AHB	CNN	0.6798	73.17%
AHB	Vanilla RNN	0.6923	71.71%
AHB	LSTM	0.6800	73.66%
AHB	GRU	0.6828	73.17%
SCP2	CNN	0.6901	55.56%
SCP2	Vanilla RNN	0.7052	53.33%
SCP2	LSTM	0.6932	51.67%
SCP2	GRU	0.7017	49.44%

7.2.2 Regression Results on Raw Signals

Table 10: Phase 2 regression performance on raw tensors

Dataset	**Model**	**Loss**	**NRMSE**
AE	CNN	0.6439	0.7990
AE	Vanilla RNN	0.4774	0.6724
AE	LSTM	0.5001	0.6876
AE	GRU	0.3852	0.6152
LFMC	CNN	1.0264	0.9221
LFMC	Vanilla RNN	1.0730	0.9342
LFMC	LSTM	1.0448	0.9197
LFMC	GRU	1.0535	0.9225

8 Comparative Discussion

8.1 Phase 1: Ridge and Dense Neural Network on Extracted Features

Phase 1 included both Ridge-based models and Dense Neural Networks, so the comparison is not only between train and test performance, but also between linear and nonlinear learning on the extracted feature space.

For classification, Ridge achieved very high training accuracy on both datasets, but its test performance dropped sharply, especially on *SelfRegulationSCP2*, where training accuracy reached 99.0% while test accuracy fell to 51.6%. Dense Neural Network behaved differently. On *AbnormalHeartBeat*, Dense NN produced 73.04% training accuracy and 73.17% test accuracy, which indicates much better generalization than Ridge and suggests that the nonlinear model avoided the strong overfitting seen in the linear baseline. On *SelfRegulationSCP2*, however, Dense NN remained at 50.00% on both train and test data, which is effectively chance-level performance and indicates underfitting or failure to extract useful decision boundaries from the provided features.

For regression, Ridge and Dense NN were closer in behavior. On *Appliances Energy*, Dense NN slightly improved training NRMSE from 0.7390 to 0.7285, but test NRMSE increased from 1.1500 to 1.2106, meaning the nonlinear model fit the training data slightly better without improving generalization. On *LiveFuelMoistureContent*, Dense NN improved both training and test NRMSE compared with Ridge, reducing the test NRMSE from 1.1000 to 1.0320. This suggests that nonlinear feature interactions were more useful on this larger regression dataset than on the smaller *Appliances Energy* dataset.

Overall, the Phase 1 comparison shows that Dense Neural Network was not uniformly better than Ridge. It clearly outperformed Ridge on *AbnormalHeartBeat* classification and on *LiveFuelMoistureContent* regression, but it underperformed on *SelfRegulationSCP2* classification and did not improve test performance on *Appliances Energy*. This indicates that the usefulness of nonlinear modeling in Phase 1 depended strongly on dataset size, feature quality, and task complexity.

8.2 Phase 2: Learning Directly from Raw Signals

The raw-signal models generally performed better than the Ridge baselines shown in Phase 1. On *AbnormalHeartBeat*, LSTM achieved the best reported accuracy at 73.66%, slightly above CNN and GRU. This suggests that long-range temporal information matters for this long ECG sequence.

On *SelfRegulationSCP2*, CNN performed best at 55.56%. This may indicate that local temporal filters were more effective than recurrent memory mechanisms for the available EEG data. Because the dataset is small and multi-channel, recurrent models may not

have had enough data to learn stable temporal dynamics.

On *Appliances Energy*, GRU achieved the best reported NRMSE of 0.6152, outperforming CNN, Vanilla RNN, LSTM, and also the Phase 1 Ridge baseline. This indicates that gated recurrence captured the short-to-medium temporal dependencies better than purely convolutional or ungated recurrent modeling.

8.3 Best and Weakest Models

Based on the available experiment summary:

- **Best classification result overall:** LSTM on *AbnormalHeartBeat* with 73.66% test accuracy.
- **Best EEG classification result:** CNN on *SelfRegulationSCP2* with 55.56% test accuracy.
- **Best reported regression result:** GRU on *Appliances Energy* with NRMSE 0.6152.
- **Weakest reported classification result:** GRU on *SelfRegulationSCP2* with 49.44% test accuracy.

8.4 Why Some Models Outperformed Others

Several dataset-specific factors explain the observed model behavior:

- **Sequence length:** *AbnormalHeartBeat* has a very long sequence length of 18,530, which makes temporal memory important. This likely helped LSTM.
- **Class imbalance:** *AbnormalHeartBeat* is imbalanced, so even a strong model may struggle without weighted loss or other imbalance-aware design.
- **Small sample size:** *Appliances Energy* has very limited training data, making overfitting a serious issue for deeper models.
- **Scaling requirements:** *SelfRegulationSCP2* and *LiveFuelMoistureContent* both require careful normalization, and poor scaling would directly harm performance.
- **Channel structure:** CNN may be more robust for local pattern extraction in multi-channel data when recurrent models are not sufficiently regularized.

9 Phase 1 vs Phase 2

The most important conclusion of the project is that learning directly from raw signals was beneficial in the reported experiments. The Ridge baselines showed strong fitting ability on training data but poor generalization. In contrast, the deep models in Phase 2 achieved better test performance on the reported datasets, suggesting that automatic feature learning from raw tensors captured information that extracted tabular features did not preserve fully.

However, the results also show that no single architecture was best for all tasks. LSTM worked best on long ECG classification, CNN worked best on the EEG classification dataset, and GRU worked best on the reported regression dataset. Therefore, model choice should depend on task type, sequence length, class balance, and data scale.

10 Conclusion

This project compared classical machine learning and deep learning techniques for ECG, EEG, and time-series classification and regression. In Phase 1, Ridge-based models were used on extracted tabular features. In Phase 2, CNN and recurrent architectures were trained directly on raw tensors. The reported results show that raw-signal learning provided better practical performance on the visible experiments, but the most effective architecture depended on the dataset. LSTM performed best for the long ECG classification task, CNN performed best for the EEG classification task, and GRU performed best for the reported regression task. Overall, the project demonstrates that data representation is just as important as model selection when working with signal data.